PRODUCT USER MANUAL

OMI_CLIMATE_SL_GLOBAL_area_averaged_anomalies
OMI_CLIMATE_SL_GLOBAL_regional_trends
OMI_CLIMATE_SL_BALTIC_area_averaged_anomalies
OMI_CLIMATE_SL_BLKSEA_area_averaged_anomalies
OMI_CLIMATE_SL_EUROPE_area_averaged_anomalie
OMI_CLIMATE_SL_IBI_area_averaged_anomalies
OMI_CLIMATE_SL_MEDSEA_area_averaged_anomalies
OMI_CLIMATE_SL_NORTHWESTSHELF_area_averaged_anomalies
Issue 6.0

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Change Record

Issue	Date	§	Description of Change	Author	Checked by	Validated by
1.0	02/02/2018	All	Creation of the document	JF Legeais		
2.0	26/07/2018	All	New supporting material, new regional MSL. Use of the reprocessed DUACS DT-2018 products with updated	JF Legeais		
2.1	25/01/2019	ALL	Evolution after review comments from MO	JF Legeais		
2.2	09/10/2019	II, III	Evolution following the update of the time series	JF Legeais		
2.3	16/01/2020	II.1, II.2	Addition of global mean sea level corrected for TOPEX-A drift.	JF Legeais		
2.4	20/08/2020	II.1	Evolution of the update frequency	JF Legeais		
3.0	09/03/2022	ALL	Use new template & temporal extension of the time series	G. Taburet	MI Pujol	C. Derval
3.1	05/04/2023	I.2	Extension of the time series	MI Pujol	MI Pujol	C. Derval
4.0	15/06/2023	I II	Use "myinteruim" datasets Nomenclature change	MI Pujol	MI Pujol	Copernicus Marine Product Management
5.0	28/02/2024	ALL	Applying global TPA correction and updated regional GIA mean to regional timeseries. Applying global TPA correction and regional GIA to map of trends. Adding global uncertainty envelope timeseries. Adding Europe region.	P. Veillard	MI. Pujol	Copernicus Marine Product Management
6.0	31/05/2025	ALL	Removing the TPA correction (no longer usable with v2024 C3S products) New temporary reference period for the computation of the GMSL parameters (trend and acceleration)	V. Quet		Copernicus Marine Product Management

GLOSSARY AND ABBREVIATIONS

CMEMS	Copernicus Marine Environment Monitoring Service
MSL	Mean Sea Level
GIA	Glacial Isostatic Adjustment
NetCDF	Network Common Data Form
OMI	Ocean Monitoring Indicator
PUM	Product User Manual
QUID	Quality Information Document

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I INTRODUCTION

I.1. Summary

This document is the user manual for the Copernicus Marine Sea Level Ocean Monitoring Indicators (OMI) including different products and associated datasets.
The products are published on the Copernicus Marine dissemination server after automatic and delayed-mode quality controls. The products are available online and disseminated through the CMEMS Information System. Files downloaded are in NetCDF format and follow CF-1.7 convention.
Disclaimer: The quality of the product may vary during the proposed time series depending on the possible update of the system.
The different sea level OMI products are described in the Quality Information Document (QUID; see References section).

I.2. History of changes

2018/02	1st publication of OMI_SEALEVEL
2018/07	New regional MSL. Use of the reprocessed DUACS DT-2018 products with updated temporal coverage.
2019/10	Temporal extension.
2020/01	Addition of global mean sea level corrected for TOPEX-A drift.
2022/03	Temporal extension.
2023/04	Temporal extension.
2023/11	Temporal extension OMIs nomenclature changed
2024/06	Applying global TPA correction and updated regional GIA mean to regional timeseries. Applying global TPA correction and regional GIA to map of trends. Adding global uncertainty envelope timeseries. Adding Europe region. Temporal extension.
2025/01/13	Remove the correction of TPA drift (no longer usable with v2024 C3S products). New temporary reference period for the computation of the GMSL parameters (trend and acceleration) without the Topex-A period.

II DESCRIPTION OF THE PRODUCT

II.1. General information

Product(s) name	OMI_CLIMATE_SL_GLOBAL_area_averaged_anomalies
Geographical coverage	Global Ocean
Variables	Time series of global area averaged sea level from satellite altimetry.
Update frequency	Seasonal
Available time series	1993-01-01 to about 5-6 months delay with present time
Temporal resolution	Daily
Integration depth (for example..)	Surface

Product(s) name	OMI_CLIMATE_SL_GLOBAL_regional_trends
Geographical coverage	Global Ocean
Variables	Map of sea level trends from satellite altimetry
Update frequency	Seasonal
Available time series	1993-01-01 to about 5-6 months delay with present time
Temporal resolution	Time integrated
Integration depth (for example..)	Surface

Product(s) name	OMI_CLIMATE_SL_BALTIC_area_averaged_anomalies
Geographical coverage	Baltic Sea
Variables	Time series of regional area averaged sea level from satellite altimetry.
Update frequency	Seasonal
Available time series	1993-01-01 to about 5-6 months delay with present time
Temporal resolution	Daily
Integration depth (for example..)	Surface

Product(s) name	OMI_CLIMATE_SL_BLKSEA_area_averaged_anomalies
Geographical coverage	Black Sea

Variables	Time series of regional area averaged sea level from satellite altimetry.
Update frequency	Seasonal
Available time series	1993-01-01 to about 5-6 months delay with present time
Temporal resolution	Daily
Integration depth (for example..)	Surface

Product(s) name	OMI_CLIMATE_SL_IBI_area_averaged_anomalies
Geographical coverage	Iberia-Biscay-Ireland
Variables	Time series of regional area averaged sea level from satellite altimetry.
Update frequency	Seasonal
Available time series	1993-01-01 to about 5-6 months delay with present time
Temporal resolution	Daily
Integration depth (for example..)	Surface

Product(s) name	OMI_CLIMATE_SL_MEDSEA_area_averaged_anomalies
Geographical coverage	Mediterranean Sea
Variables	Time series of regional area averaged sea level from satellite altimetry.
Update frequency	Seasonal
Available time series	1993-01-01 to about 5-6 months delay with present time
Temporal resolution	Daily
Integration depth (for example..)	Surface

Product(s) name	OMI_CLIMATE_SL_NORTHWESTSHELF_area_averaged_anomalies
Geographical coverage	North-West Shelf
Variables	Time series of regional area averaged sea level from satellite altimetry.
Update frequency	Seasonal
Available time series	1993-01-01 to about 5-6 months delay with present time
Temporal resolution	Daily
Integration depth (for example..)	Surface

Product(s) name	OMI_CLIMATE_SL_EUROPE_area_averaged_anomalies
Geographical coverage	Northeast Atlantic and adjacent regional seas
Variables	Time series of regional area averaged sea level from satellite altimetry.
Update frequency	Seasonal
Available time series	1993-01-01 to about 5-6 months delay with present time
Temporal resolution	Daily
Integration depth (for example..)	Surface

- All data is in NetCDF4 CF1.7 format.
- Refer to the "Data Access" section for information on data access mechanisms.

II.2. Details of the datasets

OMI_CLIMATE_SL_GLOBAL_area_averaged_anomalies Dataset : omi_climate_sl_global_area_averaged_anomalies
MSL_filtered_GIA_corrected_adjusted [m] Time series of global area averaged sea level maps from satellite altimetry. The time series is corrected from global GIA correction. It is adjusted for seasonal annual and semi-annual signals and low-pass filtered. sea_surface_height_above_sea_level
trend_MSL_filtered_GIA_corrected_adjusted [m] 2nd order least square fit on the time series. sea_surface_height_above_sea_level
uncertainty_envelop [m] Uncertainty on the time series using Quet et al., 2024 [in prep.] uncertainty budget. sea_surface_height_above_sea_level standard_error

OMI_CLIMATE_SL_GLOBAL_regional trends Dataset : omi_climate_sl_global_regional_trends
trend_GIA_corrected [mm/yr] Map of sea level trends from sea level maps from satellite altimetry. At each grid point, the trend is estimated on the time series corrected from regional GIA correction and adjusted from annual and semi-annual signals. tendency_of_sea_surface_height_above_sea_level
acceleration_GIA_corrected [mm/yr²]

Map of sea level acceleration from sea level maps from satellite altimetry. At each grid point, the acceleration is estimated on the time series corrected from regional GIA correction and adjusted from annual and semi-annual signals. -

OMI_CLIMATE_SL_BALTIC_area_averaged_anomalies Dataset : omi_climate_sl_baltic_area_averaged_anomalies
MSL_filtered_GIA_corrected_adjusted [m] Time series of Baltic Sea area averaged sea level maps from satellite altimetry. The time series is corrected from regional mean GIA correction. It is adjusted for seasonal annual and semi-annual signals and low-pass filtered. sea_surface_height_above_sea_level
trend_MSL_filtered_GIA_corrected_adjusted [m] 2nd order least square fit on the time series. sea_surface_height_above_sea_level

OMI_CLIMATE_SL_BLKSEA_area_averaged_anomalies Dataset : omi_climate_sl_blksea_area_averaged_anomalies
MSL_filtered_GIA_corrected_adjusted [m] Time series of Black Sea area averaged sea level maps from satellite altimetry. The time series is corrected from regional mean GIA correction. It is adjusted for seasonal annual and semi-annual signals and low-pass filtered. sea_surface_height_above_sea_level
trend_MSL_filtered_GIA_corrected_adjusted [m] 2nd order least square fit on the time series. sea_surface_height_above_sea_level

OMI_CLIMATE_SL_IBI_area_averaged_anomalies Dataset : omi_climate_sl_ibi_area_averaged_anomalies
MSL_filtered_GIA_corrected_adjusted [m] Time series of Iberian-Biscay-Irish Sea area averaged sea level maps from satellite altimetry. The time series is corrected from regional mean GIA correction. It is adjusted for seasonal annual and semi-annual signals and low-pass filtered. sea_surface_height_above_sea_level
trend_MSL_filtered_GIA_corrected_adjusted [m] 2nd order least square fit on the time series. sea_surface_height_above_sea_level

OMI_CLIMATE_SL_MEDSEA_area_averaged_anomalies Dataset : omi_climate_sl_medsea_area_averaged_anomalies
MSL_filtered_GIA_corrected_adjusted [m] Time series of Mediterranean Sea area averaged sea level maps from satellite altimetry. The time series is corrected from regional mean GIA correction. It is adjusted for seasonal annual and semi-annual signals and low-pass filtered.

sea_surface_height_above_sea_level
trend_MSL_filtered_GIA_corrected_adjusted [m] 2nd order least square fit on the time series. sea_surface_height_above_sea_level

OMI_CLIMATE_SL_NORTHWESTSHELF_area_averaged_anomalies Dataset : omi_climate_slnorthwestshelf_area_averaged_anomalies
MSL_filtered_GIA_corrected_adjusted [m] Time series of North-West Shelf Sea area averaged sea level maps from satellite altimetry. The time series is corrected from regional mean GIA correction. It is adjusted for seasonal annual and semi-annual signals and low-pass filtered. sea_surface_height_above_sea_level
trend_MSL_filtered_GIA_corrected_adjusted [m] 2nd order least square fit on the time series. sea_surface_height_above_sea_level

OMI_CLIMATE_SL_EUROPE_area_averaged_anomalies Dataset : omi_climate_sl_europe_area_averaged_anomalies
MSL_filtered_GIA_corrected_adjusted [m] Time series of Northeast Atlantic and adjacent regional seas area averaged sea level maps from satellite altimetry. The time series is corrected regional mean GIA correction. It is adjusted for seasonal annual and semi-annual signals and low-pass filtered. sea_surface_height_above_sea_level
trend_MSL_filtered_GIA_corrected_adjusted [m] 2nd order least square fit on the time series. sea_surface_height_above_sea_level

II.3. Product System Description

The Copernicus Marine sea level OMIs are derived from the delayed-time altimeter sea level gridded products generated by the DUACS altimeter production system and distributed by the Copernicus Climate Change Service (C3S) since these products are designed for climate applications. The DUACS altimeter processing sequences can be divided into the following main steps (details are described in Pujol et al., 2016):

- Acquisition
- Pre-processing homogenization
- Input data quality control
- Multi-mission cross-calibration
- Along-track products generation
- Gridded merged products generation
- Final quality control

Among them, the multi-mission cross-calibration process ensures that all data flows from all satellites have consistent and accurate information. This allows the reduction of various sources of geographically correlated error patterns and ensures an optimal homogenization of the dataset.
Global and regional biases between two successive altimeter missions are reduced and finally, the multi-mission along-track measurements are mapped thanks to an optimal interpolation method on a daily ¼° x ¼° regular grid. Additional information on the data processing of the altimetry data can be found in Pujol et al. (2016) and Escudier et al. (2017).

DUACS C3S maps are used to produce two types of sea level OMIs (similarly to Ablain et al., 2017).
The time series of area averaged anomalies (OMI_CLIMATE_SL_*_area_averaged_anomalies) correspond to the area average of the maps in each region weighted by the cosine of the latitude (to consider the changing area in each grid with latitude) and by the proportion of ocean in each grid (to consider the coastal areas). The time series are corrected from regional mean GIA correction (weighted GIA mean of a 27 ensemble model following Spada and Melini, 2019). The time series are no longer corrected from global Topex-A instrumental drift as it is no longer usable with new v2024 C3S products derived from the new version of Topex GDR-F products during the period 1993 – 1999. The time series are adjusted for seasonal annual and semi-annual signals and low-pass filtered at 6 months. Then, the trends/accelerations are estimated on the time series using ordinary least square fit after removing the Topex-A period from the estimation.
The map of sea level trends (OMI_CLIMATE_SL_GLOBAL_regional trends): at each grid point, the trends/accelerations are estimated on the time series corrected from regional GIA correction (GIA trend map of a 27 ensemble model following Spada and Melini, 2019) and adjusted from annual and semi-annual signals and after removing the Topex-A period.

III Data access

After registration, you will be able to download our data. To assist you, our HelpCenter is available, and more specifically its section about download.
Information on operational issues on products and services can be found on our User Notification Service. If you have any questions, please contact us.

IV FILES NOMENCLATURE AND FORMAT

Information about nomenclature of files when downloaded can be found in this article: " How is defined the nomenclature of Copernicus Marine data? | Copernicus Marine Help Center "

IV.1. Nomenclature of files

The Copernicus Marine Sea level OMI files nomenclature when downloaded through the CMEMS Services is based on the following nomenclature:

<datasetname>_<timestep>_P<productiontime>_<freetext>

Where :

- **<datasetname>** is the dataset reference.
- **<timestep>** is the first time step in the file.
- **<productiontime>** is the production time when the file was generated.
- **<freetext>** includes potential additional information.

IV.2. File Format, reading software

The products are stored using the NetCDF format.

To know more about the NetCDF format and reading software, please follow this link:
What is the format of Copernicus Marine products ? NetCDF

To understand the differences between netCDF and Zarr, please consult this article:
how-to-choose-between-netcdf-and-zarr-format-using-the-toolbox

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